



Validation Checklist

Internship Management Domain

Future Leaders Internship Platform (FLIP)

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1 Domain & Purpose

Why validation matters in the FLIP workflow

FLIP is an internship management platform connecting students and companies. Students use it to discover internship opportunities, submit applications, track their status, and build skills. Companies use it to post internship openings, review applicants, and manage the recruitment pipeline.

Validation matters because the platform sits between two parties and makes real decisions based on system data. Invalid or inconsistent data can block legitimate applicants, allow unusable internship openings, or undermine trust in application tracking.

Scope of this checklist

Three core Models are covered because they represent identity data, organizational data, and the transaction connecting them: Student, Company, and Application.

- Student — the applicant's identity and academic profile.
- Company — the employer's organizational profile.
- Application — the record linking a Student to an Internship posted by a Company, together with its status.

2 Model: Student

Represents a student user: identity, contact information, and academic affiliation.

Validators already used

Field / Rule	Validation Criteria	Consequence if it fails
nationalId	Must not be null or empty.	Registration cannot be completed without a national ID.
Email	Must not be null or empty and must be a valid email format.	Invalid email addresses may prevent account creation or communication.
Password	Must not be null or empty and must satisfy the password complexity policy (upper/lowercase, numbers, symbols).	Weak or invalid passwords may prevent registration and reduce account security.
firstName / lastName / University / studentMajor	Must not be null or empty.	Incomplete profiles prevent companies from evaluating applicants.
DOB	Must not be null and must be in the past.	Age-dependent checks may not work correctly (e.g., eligibility checks).
Phone Number	Must not be null; must start with +966 and contain 9 digits after the country code.	Registration cannot be completed without a valid phone number.
Student Overview	Maximum 200 characters.	Keeps the student overview within the defined profile limit.

Validators worth adding

- National ID format check — enforce the fixed length/digit pattern for the issuing country, not just non-emptiness or uniqueness.
- GPA range validation — enforce the valid academic scale used by the checklist (0.0–5.0) to prevent values such as –1 or 6.

Illustrative validation annotations — Student

```

@NotBlank(message = "National ID is required")
@Pattern(regexp = "^\\d{10}$", message = "National ID must contain exactly 10
digits")
private String nationalId;

@NotEmpty(message = "Email is required")
@email(message = "Invalid email format")
private String email;

@NotBlank(message = "Password is required")
@Pattern(regexp = "^(?=.*[a-z])(?=.*[A-Z])(?=.*\\d)(?=.*[@$!%*?&]).{8,}$",
        message = "Password must contain at least 8 characters, one uppercase
letter, one lowercase letter, one number, and one symbol")
private String password;

@NotBlank(message = "First name is required")
private String firstName;

@NotBlank(message = "Last name is required")
private String lastName;

@NotBlank(message = "University is required")
private String university;

@NotBlank(message = "Major is required")
private String studentMajor;

@NotNull(message = "Date of birth is required")
@Past(message = "Date of birth must be in the past")
private LocalDate dateOfBirth;

@NotBlank(message = "Phone number is required")
@Pattern(regexp = "^\\+9665\\d{8}$", message = "Phone number must be a valid +966
mobile number")
private String phoneNumber;

@NotNull(message = "GPA is required")
@DecimalMin(value = "0.0", message = "GPA cannot be less than 0")
@DecimalMax(value = "5.0", message = "GPA cannot be more than 5")
private Double gpa;

@Size(max = 200, message = "Maximum 200 characters")
private String overview;

```

3 Model: Company

Represents an employer's organizational profile, used to post internships and manage applicants.

Validators already used

Field / Rule	Validation Criteria	Consequence if it fails
Email	Must not be null or empty.	Helps prevent duplicate or conflicting company accounts/postings.
Password	Must not be null or empty and must satisfy the password complexity policy.	Weak credentials can expose applicant data such as resumes and contact information.
CompanyName / Location	Must not be null or empty and must respect the defined character limit.	Students cannot meaningfully evaluate or search for a company with missing core information.
Industry	Must not be null and must be one of the predefined industry values.	The request is rejected if the industry is missing or invalid.
Application Start Date	Cannot be null; must be present or future.	Prevents misleading or unusable application windows.
Application End Date	Cannot be null; must be in the future.	Prevents misleading or unusable application windows.

Validators worth adding

- Website format check — validate that the website field contains a valid URL rather than unconstrained text.
- Duplicate-company detection — compare company name plus website/domain at signup to catch accidental double registration.
- Company status verification — requires the company status to be Active before it can post internships or collect student documents.
- Date consistency check — ensure the internship end date is not earlier than its start date.

Illustrative validation annotations — Company

```

@NotBlank(message = "Email is required")
@email(message = "Invalid email format")
private String email;

@NotBlank(message = "Password is required")
@Pattern(regexp = "^(?=.*[a-z])(?=.*[A-Z])(?=.*\\d)(?=.*[@$!%*?&]).{8,}$",
        message = "Password must contain at least 8 characters, one uppercase
        letter, one lowercase letter, one number, and one symbol")
private String password;

@NotBlank(message = "Company name is required")
@Size(min = 2, max = 20, message = "Company name must be between 2 and 20
characters")
private String companyName;

@NotEmpty(message = "Location is required")
@Size(max = 20, message = "Location must not exceed 20 characters")
private String location;

@NotNull(message = "Industry is required")
@Pattern( regexp = "TECHNOLOGY|GAS|ENGINEERING")
private String industry;

@NotNull(message = "Start date is required")
@FutureOrPresent(message = "Start date must be today or in the future")
@JsonFormat(pattern = "yyyy-MM-dd")
private LocalDate startDate;

@NotNull(message = "End date is required")
@Future(message = "End date must be in the future")
@JsonFormat(pattern = "yyyy-MM-dd")
private LocalDate endDate;

```

4 Model: Application

The transactional record linking a Student, a Company, and the Internship they applied to, plus its status.

Validators already used

Field / Rule	Validation Criteria	Consequence if it fails
InternshipId / StudentId / CompanyId	Must not be null	The application is rejected because it would reference missing or invalid records.
Status	Must be one of the predefined values: Pending, Under Review, Accepted, or Rejected.	An invalid status could break application tracking and dashboard filters.
ApplicationDate	Must not be null and should be automatically set to the current timestamp when the application is created.	The application cannot be properly tracked in the timeline.
PositionDescription	Must not be blank and should have a maximum length.	The application is rejected because the position value would be missing or invalid.
Application Status Transition	Status changes must follow an allowed sequence, such as Pending → Under Review → Accepted or Rejected.	The update is rejected if it attempts an invalid transition, such as Rejected → Pending.

Validators worth adding

- DecisionDate must be later than ApplicationDate: prevents a decision timestamp that predates the application itself.
- Deadline enforcement — block new applications after the internship's application deadline has passed.
- Capacity enforcement — the number of accepted applications must not exceed the number of available internship positions.

Illustrative validation annotations — Application

```
@NotNull(message = "Internship ID is required")
private Integer internshipId;

@NotNull(message = "Student ID is required")
private Integer studentId;

@NotNull(message = "Company ID is required")
private Integer companyId;

@NotNull(message = "Status is required")
@Pattern(regexp = "pending|underreview|accepted|rejected")
private String status;

@NotNull(message = "Application date is required")
@JsonFormat(pattern = "yyyy-MM-dd")
private LocalDate applicationDate;

@NotBlank(message = "Position description is required")
@Size(min = 2, max = 100, message = "Description must be between 2 and 100 characters")
private String description;
```